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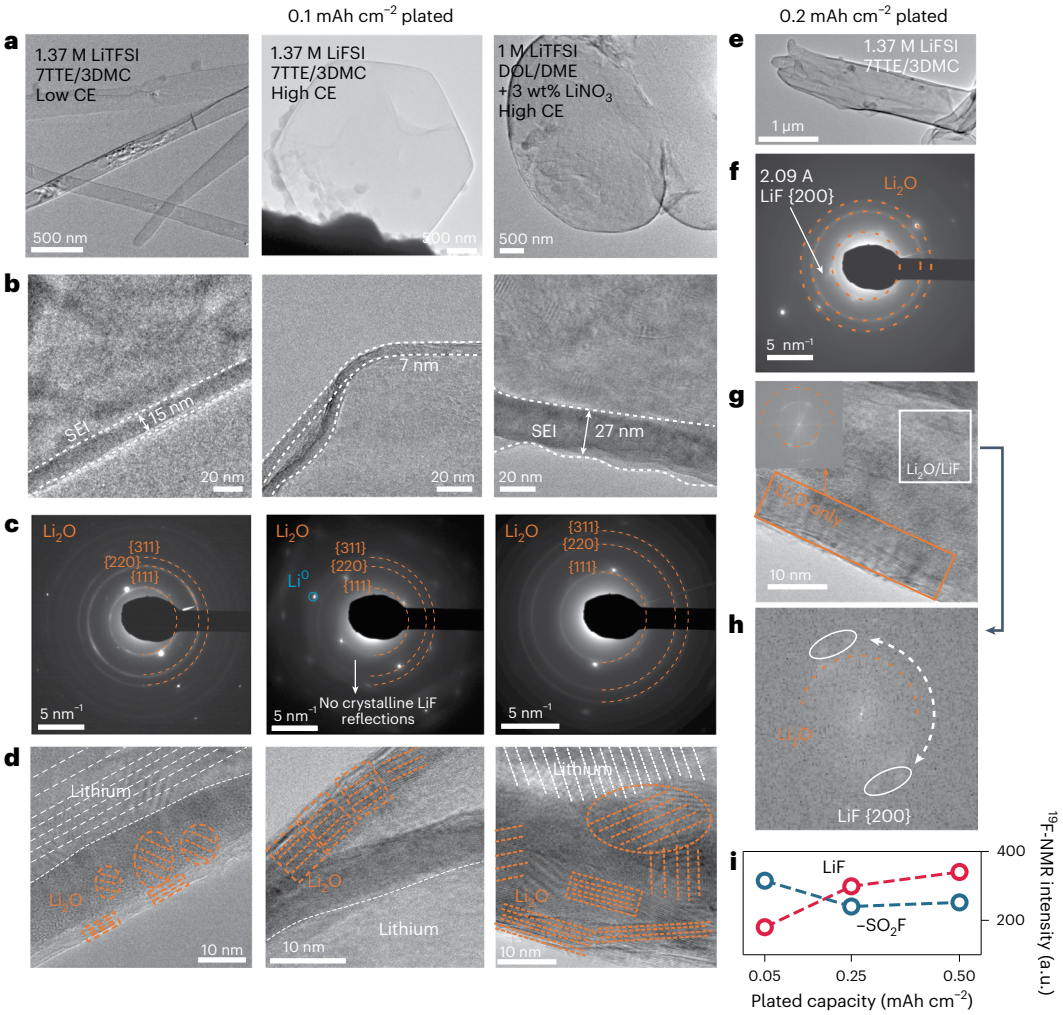

Fig. 4 | Cryo-TEM imaging of Li_2O in plated Li electrodes. a, Low magnification images of the plated Li deposit morphology (0.1 mAh cm^{-2}, no stripping) in low- and high-CE electrolytes, chosen from the quantification analysis in Fig. 2. **b**, Magnification of the interface between the Li deposit and vacuum, highlighting the thickness of the SEI. **c**, SAED measurements near the interface. Enlarged and non-annotated versions are shown in Supplementary Fig. 50. **d**, HRTEM images of the crystalline lattices within the SEI. **e**, Low magnification image of a Li deposit in the 1.37 M LiFSI 7TTE/3DMC LHCE electrolyte, now at 0.2 mAh cm^{-2} plated capacity. **f**, SAED patterns of the SEI. **g**, HRTEM images of the crystalline phases in the higher-capacity SEI. Inner frame indicates the Fourier transform of the outer SEI region with only Li_2O reflections. **h**, Fourier transform of the white region indicated in **g**, with both Li_2O and LiF reflections. **i**, ^{19}F-NMR measurements of -SO_2F and LiF in plated Li electrodes.

to 1.00 M, CE increases to >97%, coinciding with a notable increase in Li_2O and, more modestly, LiF (Fig. 5c). At 1.25–1.37 M, a substantial increase in -SO_2F is seen by ^{19}F-NMR and S in ICP, along with a suppression in CF_x fragments from TTE decomposition (Fig. 5d and Supplementary Fig. 46), thus providing conclusive evidence that LiF derives primarily from anions in this regime. Li_2O can, in principle, form from DMC, though a more plausible pathway involves fragmentation of LiFSI[54], which is in accordance with increased contact-ion pairing and appearance of other salt byproducts (S, LiF, -SO_2F). However, relatively little Li_3N was observed, indicating that complete anion fragmentation and incorporation of N as a fully reduced phase occurs less readily than LiF/Li_2O formation. These results reveal that $LiFSI^-$-based LHCEs benefit from their ability to promote formation of Li_2O as a key ionic phase resulting from elevated anion reactivity, especially at FSI^- concentrations >0.75 M (Supplementary Fig. 45b,c) with correspondingly high CE. If true, other salts forming Li_2O could in principle perform an equivalent function without the coupled formation of LiF, relaxing sole reliance on LiFSI for high Li CE.

To test this hypothesis, we designed an additional series of electrolytes to modulate Li_2O and LiF proportions beyond those achievable with LiFSI (Fig. 6). The recipe for LHCE preparation described by Ren et al.[38] was adapted, starting with a fixed 3:1 molar ratio of diluent:salt and adding DME until full salt dissolution, to produce LHCEs with selective F or O enrichment. TTE/DME electrolytes[37,38,52] with fluorinated, O-free salts (1.52 M $LiBF_4$ or 1.57 M $LiPF_6$) yielded poor CE (~ 65% and ~91%, respectively; Fig. 6a,b), low Li_2O (Fig. 6c) and greater amounts of LiF. On the other hand, an analogous LHCE based on oxygenated $LiClO_4$, that is, an F-free salt, achieved higher CE (98.7%) and formed a larger amount of Li_2O (Fig. 6c). To completely exclude any source of fluorine from the electrolyte, TTE was replaced with anisole[55], forming an entirely F-free electrolyte, which led to a slightly higher CE (98.9%). This 'defluorination' strategy was then applied to the 1 M LiTFSI DOL/DME + 3 wt% $LiNO_3$ tested previously in Figs. 2–4 but now replacing the LiTFSI with $LiClO_4$, yielding yet another fluorine-free electrolyte with high CE (99.1%). Accordingly, addition of the Li_2O-forming additive to the $LiClO_4$ TTE/DME LHCE electrolyte also further increased its CE to 99.1% CE. These results indicate that electrolytes yielding Li_2O-rich, LiF-free interphases represent a promising alternative to LiFSI-based LHCEs for high-performance Li anodes; such LiFSI-based electrolytes were measured here to have CE = ~99.3% (Supplementary Fig. 47) but are

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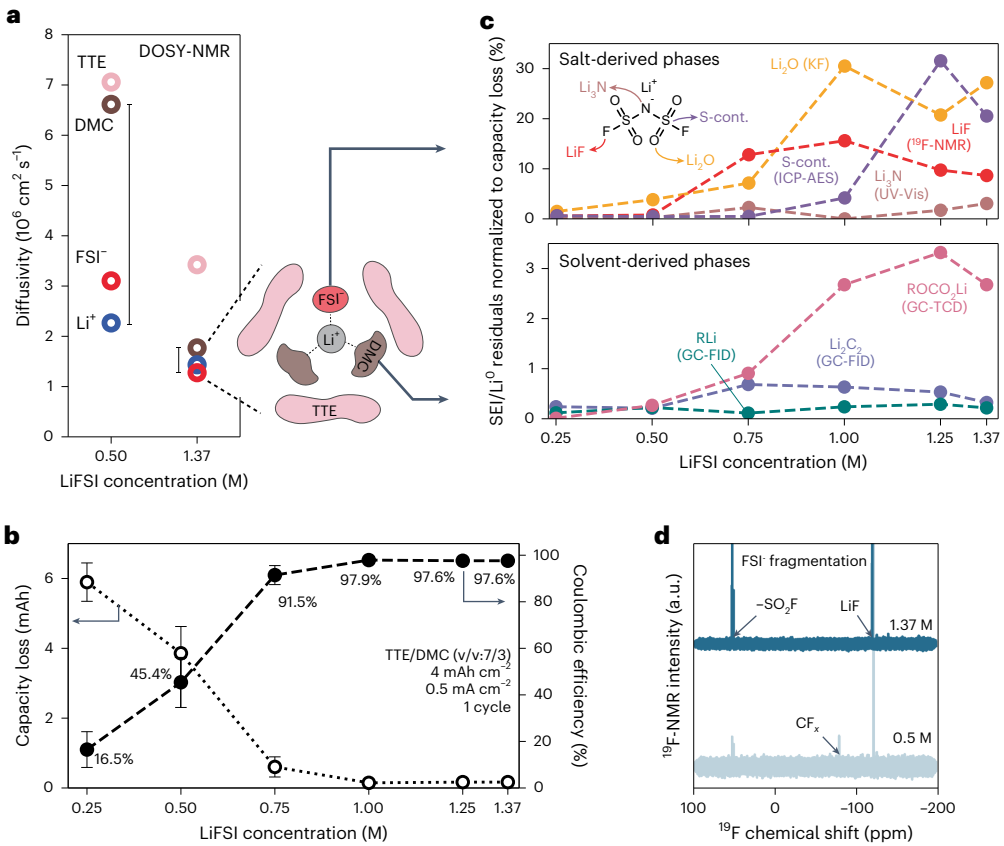

Fig. 5 | Li_2O formation in LiFSI-based LHCE electrolytes. a, Diffusivity of all electrolyte components in 0.5 M and 1.37 M LiFSI TTE/DMC measured by DOSY-NMR, revealing the formation of DMC-Li^+-FSI^- coordination shells. **b**, Capacity loss and CE after the first cycle (4 mAh cm^{-2}, 0.5 mA cm^{-2}) in the LHCE-like electrolytes LiFSI 7TTE/3DMC, with salt concentrations increasing from 0.5 M to 1.37 M. Data presented as mean ± standard deviation (n = 3 per concentration). **c**, Salt-derived SEI phases (Li_3N, S-containing, LiF and Li_2O) and solvent-derived SEI phases ($ROCO_2Li$, Li_2C_2, RLi) normalized to capacity loss, measured after the first cycle. **d**, ^{19}F-NMR of electrode residuals dissolved in H_2O after the first cycle in the 0.5 M and 1.37 M electrolytes.

often corrosive and toxic due to their high fluorine content. Whereas the particular blends reported herein are not optimal for full cells due the instability of non-fluorinated ethers at cathode potentials (Supplementary Fig. 48), safety concerns associated with strongly oxidizing salts such as $LiClO_4$[24], and limited high rate performance due to solvent viscosity (Supplementary Fig. 49), these challenges motivate further research on non-fluorinated but cathodically stable salts and solvents such as sulfates and carbonates[24].

Thus, despite LiF being widely regarded as the major desirable SEI phase, LiF is not required to achieve >99% CE with Li anodes, as demonstrated by the designed oxygen-rich, fluorine-free electrolytes (Fig. 6). Such observation is not true for Li_2O: not only did Li_2O show a stronger correlation with CE than LiF (Fig. 3), but no examined electrolyte in this study could achieve such CEs without having Li_2O as a major SEI phase. These findings highlight Li_2O content as a powerful SEI-focused descriptor to guide broad electrolyte discovery, contrasting with existing electrolyte-focused descriptors[15,25,26,52,56–59] that are effective when utilizing LiFSI salts, but which have not yet effectively described performance across broader electrolyte classes. Nonetheless, continued efforts are needed to understand intriguing questions raised by these results. These include whether and which electrolyte species, beyond anions, may source Li_2O and the conditions that promote or hinder such Li_2O-forming pathways to enable finer control over Li_2O structure in the SEI. It is also critical to examine the role of phases beyond Li_2O and LiF as noted herein; and insofar invisible SEI phases (for example, alkoxides, oligo/polymeric phases), as their possible contribution to high CE cannot yet be commented upon. Similarly, the results cannot exclude the significance of non-LiF fluorinated phases, which were detected by ^{19}F-NMR. For example, when LiFSI is used as a salt, both Li_2O- and SO_2F-containing phases were seen to increase with CE and salt concentration, which indicates that there may be a relationship between these phases. Quantification of their effect on CE, along with needed advances in the chemical resolution and quantification of anion fragments more broadly, may be possible in the future through multidimensional NMR experiments and represents a frontier in improving SEI chemical resolution. Furthermore, the titration scheme presented herein is not limited to Li and can be extended to anode chemistries such as graphite and Na for which titration methods have started to be developed[30,60] but which currently lack quantitative methods for oxide and fluoride characterization. Finally, we emphasize that electrolyte fluorination is still currently necessary for full Li-ion cells (Supplementary Fig. 48)[14], as it aides the cathode by preventing Al corrosion and promoting oxidative stability. Future efforts should focus on integrating and balancing beneficial fluorination and oxygenation features to achieve both anodic and cathodic improvements while balancing parallel demands regarding corrosion, viscosity/wettability, cost and sustainability.

Conclusions

A titration methodology was developed to precisely measure total Li_2O content in cycled Li anodes, adding to a growing suite of chemical titrations available for probing Li SEI composition encompassing LiF, Li_3N, $ROCO_2Li$, Li_2C_2, RLi, P-, S- and B-containing phases in addition to inactive Li^0 formed during cycling. Among these, Li_2O was found to be the most prevalent phase at high CE across a chemically diverse set of electrolytes. Consistently, statistical analyses on the titration data

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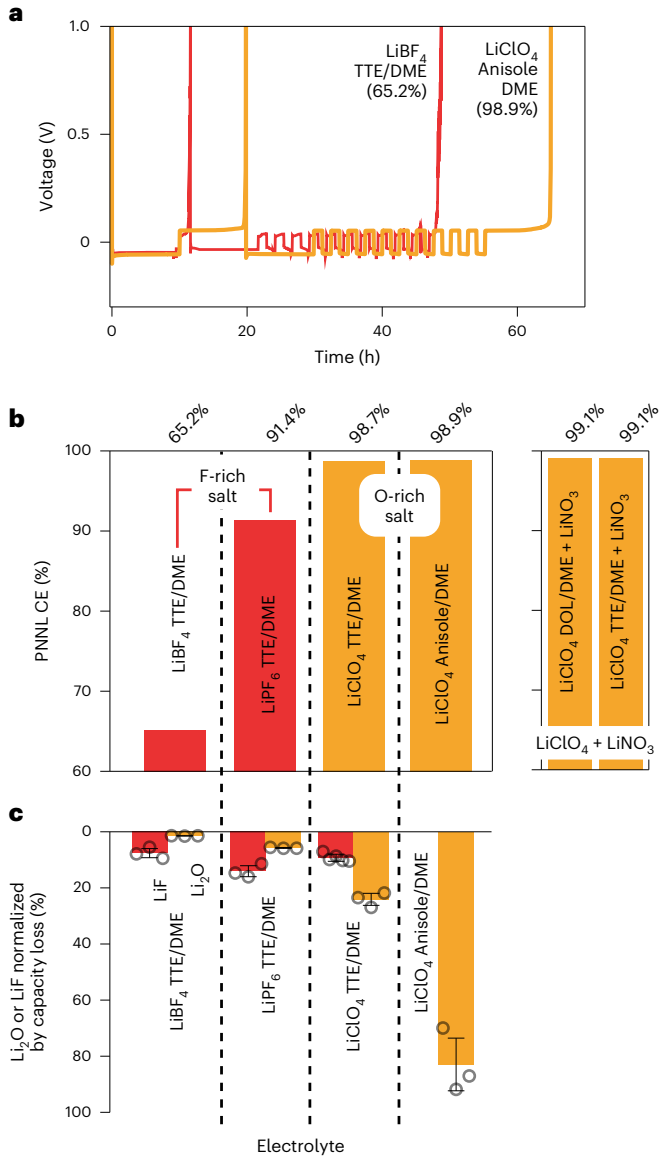

Fig. 6 | Design of fluorine-free high-CE electrolytes. a, Cycling data used to measure CE in the LHCEs tested herein (0.4 mA cm^{-2}, 4 mAh cm^{-2} formation cycle, 4 mAh cm^{-2} reservoir, 0.5 mAh cm^{-2} cycles), following the Pacific Northwestern National Laboratory (PNNL) protocol[61]. **b**, Summary of CEs measured for LHCEs containing $LiBF_4$, $LiPF_6$, $LiClO_4$ and $LiNO_3$ as salts. **c**, Total amounts of Li_2O and LiF normalized by capacity loss in select electrolytes. In **b**–**c**, full electrolyte formulations correspond to, in order of CE: 1.52 M $LiBF_4$ 0.69-TTE/0.31-DME; 1.57 M $LiPF_6$ 0.71-TTE/0.29-DME; 1.7 M $LiClO_4$ 0.77-TTE/0.23-DME; 2.27 M $LiClO_4$ 0.74-anisole/0.26-DME; 1 M $LiClO_4$ DOL/DME + 3 wt% $LiNO_3$; 1.7 M $LiClO_4$ 0.77-TTE/0.23-DME + ~0.1 M $LiNO_3$; solvent fractions given in vol/vol. Data presented as mean ± standard deviation, and open markers denote individual cells (n = 3 for all phases, except n = 5 for LiF in $LiClO_4$ TTE/DME).

revealed that Li_2O has the strongest positive correlation with CE with a correlation coefficient of $\rho > 0.9$, even exceeding that of the broadly regarded LiF ($\rho = 0.758$), countering the prevailing wisdom that LiF is a uniquely important SEI building block. Cryo-TEM analyses showed the importance of the morphology of Li_2O nanostructures, where in low-CE electrolytes Li_2O forms with seemingly arbitrary crystal planes, contrasting with the highly organized particles found at high CE that are oriented along the SEI interface. LiFSI, the current leading salt in modern electrolyte design, was shown to be a major driver of Li_2O formation, especially in systems where contact-ion pairing is promoted. Finally, the critical role of Li_2O vs LiF was leveraged to demonstrate that completely fluorine-free, oxygen-rich electrolytes can breach 99% Li CE. This insofar-neglected strategy of SEI oxygenation opens an unexplored design space for high-CE electrolytes, enabling an alternative route complementary to, and potentially competitive with, fluorination. Such a route is becoming increasingly important as industry seeks to move beyond costly, toxic fluorination schemes that governed the past decades of electrolyte development towards more sustainable paradigms for future battery designs.

Methods

Cell assembly and cycling

Coin cell components (CR2032, MTI) were cleaned with ethanol and deionized (DI) water, then dried in the vacuum oven at 70 °C for at least 12 hours and taken inside an Ar glovebox (MBraun, <0.1 ppm H_2O, <0.1 ppm O_2) without ambient exposure. Cu foil current collectors (15 mm diameter) were soaked in 1 M HCl for 1 hour, rinsed with DI water and similarly dried in a vacuum glass oven at 70 °C. Polymeric separators (typically Celgard 2325; 3501 used for 1 M $LiClO_4$ PC due to electrolyte wetting) were punched to 20 mm and were similarly dried and transferred. Lithium foil discs (Alfa Aesar) were punched to 15 mm inside of the glovebox. Each cell was prepared with a total of 50 µl of electrolyte (25 µl added before each separator). After assembly, cells were sealed using an automatic crimper (MTI, 0.82 T load setting). The assembled cells were then taken outside of the glovebox, rested for 5 hours at open circuit voltage (OCV) and cycled galvanostatically at 0.5 mA cm^{-2} at room temperature in a battery cycler (BCS, MTI) for a number of cycles determined by the target capacity loss amounts (Supplementary Note 4), with a stripping cut-off voltage of 1 V. Between each plating or stripping half cycle, the cells rested for 5 minutes at OCV. Coulombic efficiency was determined as the average cycling efficiency over all galvanostatic cycles, unless explicitly noted otherwise (Fig. 6).

Sample preparation and titration

After cycling, cells were taken inside the glovebox and unsealed using an automated decrimper (MTI). The cycled Cu current collector containing SEI/Li^0 residuals was carefully extracted from the cell and prepared according to the type of reactant used for titration. For acid/water titration, samples were soaked in anhydrous DME or DMC for 3 min before the reaction to remove excess electrolyte (Supplementary Fig. 9), dried under antechamber vacuum and taken outside of the glovebox in gas-tight vials. For Li_2O quantification, 2-butoxyethanol was used for titration, and no rinsing was required. For LiF, P-containing phases and Li_3N quantification, H_2O was used for titration. For Li^0, $ROCO_2Li$, Li_2C_2, RLi, S-containing and B-containing phases, 3.5 M HCl was used for titration. Supplementary Methods provide detailed information on each analytical technique, their appropriate calibrations and sensitivities.

For HCl titration, 500 µl of 3.5 M HCl was injected into the vial using a gas-tight syringe. After >8 hours, 2.5 ml of gas was extracted from the headspace of the vial and injected into a GC (Agilent 7890 A) to quantify the amounts of H_2, CO_2, C_2H_2, CH_4, C_2H_6, C_2H_4, which determine the total amount of Li^0, $ROCO_2Li$, Li_2C_2 and RLi. The vial was opened and another 500 µl of DI water was added, further diluting the liquid solution in a total volume of 1 ml. Using the resulting solution, samples were then prepared for ICP analysis to determine the concentration of S- and B-containing phases in solution. The acid titration scheme is depicted in Supplementary Fig. 6.

For H_2O titration, 500 µl of DI H_2O was injected into the vial using a gas-tight syringe. After >8 hours, 2.5 ml of gas was extracted from the headspace of the vial using a gas-tight syringe and injected into the GC for analysis. Then, the vial was opened and another 1 ml of DI water was added, further diluting the liquid solution in a total volume of 1.5 ml. Using the resulting solution, samples were then prepared for ^{19}F-NMR to determine the concentration of LiF; for ICP-AES to determine the concentration of P-containing phases; and for UV-Vis analysis